

Project Status Update: Roof Ventilation & Insulation Retrofit

Date: January 11, 2026 **Property:** 1517 San Juan Ave

Status: CRITICAL PAUSE (Insulation Work Suspended)

1. Executive Summary

The insulation retrofit (Advance Insulation) is officially paused. New "thermal tracking" (ghosting) and mold in a second bedroom confirm active air leakage from the interior to the attic.

Forensic calculation identifies the root cause as a severe **Ventilation Imbalance** from the 2024 roof replacement. The attic is depressurized, acting as a vacuum on the living space.

2. The Technical Failure (The Math)

- **Exhaust Capacity (Ridge):** $60 \text{ ft Omni Vent} \times 18 \text{ NFA/ft} = \mathbf{1,080 \text{ sq. in.}}$
- **Intake Capacity (Eave):** $12 \text{ SV-50 Vents} \times 50 \text{ NFA/unit} = \mathbf{600 \text{ sq. in.}}$ (Theoretical Max)
- **The Deficit:** $\mathbf{-480 \text{ sq. in.}}$ (Minimum)
- **Result:** The ridge vent is starving for air, creating negative pressure that pulls warm, moist air through the ceiling plane.

3. Recent Developments (Jan 10-11, 2026)

A. The Insulation Pause

- **Action:** Notified Brodie (Advance Insulation) to hold scheduling.
- **Reason:** Confirmed that insulating before fixing the intake would trap moisture and fail to stop the vacuum effect.

B. The Text Exchange with Vintage Roofing (Mark)

- **User Action:** Informed Mark of the calculation (1080 vs 600) and the plan to hire independent inspectors.
- **Mark's Admission:** *"If these soffits were to get converted / vented. Those low air vents would alter the air flow by exhaust instead of drawing air."*
- **Strategic Win:** Mark explicitly admitted that his vents (SV-50) are incompatible with proper soffit ventilation. This confirms they **must be removed** if soffits are upgraded.

4. The Action Plan (The "Split Solution")

The remediation strategy separates the "Correction" from the "Upgrade" to minimize conflict and ensure quality.

Phase 1: Verification (Current Stage)

- **Objective:** Obtain a written/verbal "Failure Diagnosis" from a neutral 3rd party.
- **Target Companies:** **Parker Johnston, Proline, or CSC** (for inspection/report only).
- **Key Question:** *"Can these 12 SV-50s mathematically support a 60-foot Omni Ridge Vent?"*
- **Status:** Scheduling in progress.

Phase 2: The Demand (Warranty Claim)

- **Trigger:** Immediately following the independent inspection.
- **The Offer to Mark:**
 1. **Mark (Vintage):** Removes 12 SV-50 vents and patches the roof deck (Warranty Work).
 2. **User (You):** Hires a specialist to cut continuous soffit vents (The Upgrade).

Phase 3: Execution

- **Step 1 (The Upgrade):** **Irwin Industries (or similar Building Envelope Specialist)** cuts 3" continuous strip vent in stucco (~1,080+ sq. in. intake).
- **Step 2 (The Repair):** Vintage Roofing patches the 12 roof deck holes.
- **Step 3 (The Restoration):** Advance Insulation performs mold remediation, air sealing, and insulation install once attic pressure is neutral.

5. Outstanding Questions / Risks

- **Mark's Resistance:** Will he try to argue the vacuum effect isn't his fault?
 - *Counter:* The math (1080 vs 600) and Manufacturer Manual ("Not for Intake") override his opinion.
- **Inspection Timing:** Need to secure an inspection date ASAP to prevent long delays.